

Syllabus/Course Info

Welcome to Math 411!

We will work together for a few weeks to study basic properties of the integers, the polynomial rings, and some related ring extensions. Those are arguably the most fundamental object in the study of algebra, and a solid grasp of the material may allow you move to more advanced topics, such as algebraic number theory, Galois theory or even algebraic geometry. I hope we can have a nice time together and all learn something we didn't know before!

Instructor: **Eric Zhang** zhangj20@uw.edu (<mailto:zhangj20@uw.edu>)

Office: **Padelford Hall C-8-M**

Classroom: **MWF 8:30 - 9:30 AM @ Paccar Hall** 
(<https://www.washington.edu/classroom/PCAR>) **293**

Office hour: **M 3:00 - 4:00 AM @ Paccar Hall**  (<https://www.washington.edu/classroom/PCAR>)
293

W 3:00 - 4:00 AM on Zoom (920 7327 5444)

Extra Zoom office hour before the exam: 9-10 pm on July 20, 22 and 8-10 pm on Aug 19, 20

Course content: We will work together to study the basic properties of integers and some ring extensions of integers. The sequel Math 412 will be devoted to the study of the basic properties of the polynomial rings. In particular, for Math 411, we will talk about divisibility, primes, congruence, the Fundamental theorem of Arithmetic, Fermat's little theorem, Euler's theorem, the Chinese remainder theorem, and general ring properties such as being UFD or PID etc.

Textbook: *Integers, Polynomials and Rings* by Ron Iravings

Homework: There are weekly homework assigned each Friday and is due next Friday. You can and are encouraged to work in groups but you must write the solutions by yourself. It should be submitted through Gradescope, preferably in a pdf file generated by Latex. **Request for extension or late submission must be sent at least by Wednesday night.** *Late homework submitted on Saturday, Sunday, Monday etc. will be penalized in the way that the maximum grade will be 90%, 80%, 70%, etc.*

Exams: There will be **two one-hour, in-class midterm exams**, on the last day of each term. That is,

Midterm 1 on July 23 from 8:30 to 9:30 AM,

Midterm 2 on August 22 from 8:30 to 9:30 AM.

If you will not be able to attend the exam in person, you need to take it on an acceptable alternative proctoring center following the instructions on this page

<https://math.washington.edu/exam-administration> → <https://math.washington.edu/exam-administration>).

Grades: 10% Participation + 30% of Homework + 30% of Midterm 1 + 30% of Midterm 2

Schedule

Week/Day	Monday	Wednesday	Friday
Week 1 (Jun 23 - 29)	Introduction Division Algorithm	Greatest Common Divisor Euclidean Algorithm	Congruence Homework 1
Week 2 (Jun 30 - Jul 6)	Primes numbers the FTA	Review I: Properties of Integes	Hoilday Homework 2
Week 3 (Jul 7 - 13)	Groups and Rings $\mathbb{Z}[\sqrt{2}]$	More on Groups and Rings	Unit Groups: $(\mathbb{Z}_m^\times, \times)$ Homework 3
Week 4 (Jul 14 - 20)	Group order, Lagrange's theorem	Ideals: prime ideal, maximal ideal	Domains: ED, PID, UFD Homework 4
Week 5 (July 21 - 27)	Review II: Properties of Groups/Rings	Midterm 1	Polynomial ring over a field Homework 5
Week 6 (July 28 - Aug 3)	Polynomial ring over a field is Euclidean domain	Roots of polynomials over a field	Polynomial ring over a commutative ring Homework 6
Week 7 (Aug 4 - 10)	$\mathbb{Z}[x]$ and	Gauss's Lemma	Eisenstein Criterion

	Primitive polynomial	irreducibles in $\mathbb{Z}[x]$	Homework 7
Week 8 (Aug 11 - 17)	irreducibles in $\mathbb{Z}_p[x]$ Reduction mod p	Reduction mod p Polynomial congruence	Polynomial congruence Homework 8
Week 9 (Aug 18 - 24)	Review III: Ideals, irreducibles in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$	Review IV: homeworks, practice exam, midterm	Midterm 2

Accommodations:

I fully support necessary accommodation, requests, or suggestions in general based on religious, physical, mental, or any other reasons. If you have any concerns along the course, please tell me and we will work something out.

Religious Accommodations:

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy \(/staff-faculty/religious-accommodations-policy/\)](https://registrar.washington.edu/staff-faculty/religious-accommodations-policy/)  (<https://registrar.washington.edu/staff-faculty/religious-accommodations-policy/>). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form \(/students/religious-accommodations-request/\)](https://registrar.washington.edu/students/religious-accommodations-request/)  (<https://registrar.washington.edu/students/religious-accommodations-request/>).

Disability Accommodations:

Disability accommodations or Disability resource can be acquired from the DRS office <https://hr.uw.edu/dso/services/> (<https://hr.uw.edu/dso/services/>).